INCH-POUND
MIL-L-70847 (AR)
23 January 1992

MILITARY SPECIFICATION

LASER TUBE/MODULATOR ASSEMBLIES, 105MM MAIN GUN, 105MM AND 152MM COAX, GENERAL SPECIFICATION FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture, inspection, packaging and acceptance of the following Laser Tube/Modulator Assemblies: 152mm COAX, 11835693; 105mm Main Gun, 11749644 and 105mm COAX, 11835692.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Comander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1265

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel Manufacture and Inspection of, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-W-63150	-	Weapons and Support Materiel, Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices)
MIL-STD-1695	-	Environments, Working, Minimum Standards for
MIL-STD-2000	-	Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-45662	-	Calibration System's Requirements

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HANDBOOK

MILITARY

DoD-HBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices) METRIC

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER.

PRODUCT DRAWINGS (CAGE CODE 19200)

11749188	-	Printed Wiring Board Assembly, 105mm (152mm)
PL11749188-1	-	Printed Wiring Board Assembly, COAX
PL11749188-2	-	Printed Wiring Board Assembly, 105mm Main Gun
11749308	-	Schematic Diagram, 105mm, 152mm, COAX Transmitter
11749048	-	Laser Tube Assembly
11749205	-	Schematic Diagram, Laser
11835693	-	Laser Tube/Modulator Assy, 152mm COAX
11749644	-	Laser Tube/PWBA, 105mm Main Gun
11835692	-	Laser Tube/Modulator Assy, 105mm COAX

INSPECTION EQUIPMENT (CAGE CODE 19200)

11748854	-	Detector Module, Si-PN (see 6.8)
11749020	-	Printed Wiring Board Assembly, Helmet (MWLD) (see 6.8)

PACKAGING DATA SHEET (CAGE CODE 19200)

SPI-P11835693	-	Laser Tube/Modulator Assy, 152mm COAX
SPI-P11749644	-	Laser Tube/PWBA, 105mm Main Gun
SPI-P11835692	-	Laser Tube/Modulator Assy, 105mm COAX

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(Copies of other Government documents, drawings and publications required by the contractor in connection with specific acquisition funtions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The laser tube/modulator assemblies include a laser tube assembly and a modulator board. The modulator board contains the custom laser driver integrated circuit (IC) chip, switch transistors, modulator transformer, temperature compensation components (RT 1 and RT 2 and R3-R7), and current sense resistors (R8 and R9).

The integrated circuit (IC) receives load and fire pulses from remotely located CMOS circuitry. The load pulses turn on the proper switch transistor which is then turned OFF as the result of a comparison of the voltage drop across the current sense resistor with the internal reference within the laser driver chip.

At turnoff, the stored energy in the transformer is transferred to the energy storage capacitor located in the laser tube assembly.

The subsequent fire pulse is fed to the gate of the SCR, also located in the laser tube assembly. This causes the energy storage capacitor to discharge through the laser diode generating an optical pulse of controlled energy by virtue of the control of energy storage in the transformer provided by the current limit comparison described above.

The laser tubes and modulator assemblies are configured in accordance with Table I.

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TABLE I. Laser Tube/Modulators Configurations.

System	Modulator	Laser Tube
105mm and 152mm COAX	11749188-1	11749048-9
105mm Main Gun	11749188-2	11749048-8

3.1.1 Matched assemblies. The laser tube and modulator assemblies in the above configurations (see Table I) shall be matched by selecting R8 and R9 (see Figure 1) such that the near miss energy and irradiance energy for the systems specified are met (see Table II).

TABLE II. Resistor Selection

System	R8 (ohms)	R9 (ohms)	Near Miss Energy	Irradiance Energy
105mm and 152mm COAX	4.75 - 24.9	1.1- 15.0	3.4.2.2	3.4.2.3
105mm Main Gun	2.21- 8.25	0.511- 3.01	3.4.2.4	3.4.2.5

3.2 First article. When specified (see 6.4), a sample shall be subjected to first article inspection in accordance with 4.4.

3.3 Environmental stress screen (ESS). All assemblies shall be subjected to an environmental stress screen prior to conformal coating consisting of temperature cycling and soaking between the temperatures -35°c and +62.8°C. After exposure to ESS, the characteristics of 3.4 shall not be degraded.

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3.4 Characteristics.

3.4.1 Operating conditions. The operating conditions shall be in accordance with Table III, measured under quiescent conditions (unit not firing). For terminal and component designations called out in this specification, see 11749205, Schematic Diagram, Laser and 11749308, Schematic Diagram, 105mm, 152mm, COAX Transmitter.

TABLE III. Operating Conditions.

Parameter	Terminal (of the PWBA)	Minimum	Nominal	Maximum	Unit
Ambient Temperature		-31.7	25	62.8	°C
Input (+) Power	3,4	7.0	9.0	12.0	Vdc
Return (-) Power	8,9,10, 13,14,15				
Input Current	3,4		2.0	3.0 1/	mA

NOTE:
1/ Quiescent condition.

3.4.2 Performance. Unless otherwise specified, the laser tube/modulator assembly shall meet the requirements of this specification under the following conditions:

a.	Temperature	25°±10°C (77°±18°F)
b.	Relative humidity	50±30 %
c.	Barometric pressure	725+50, -75mm Hg (28.5+2.0, -3.0 inches Hg)
d.	Input power and loads	See Figure 1

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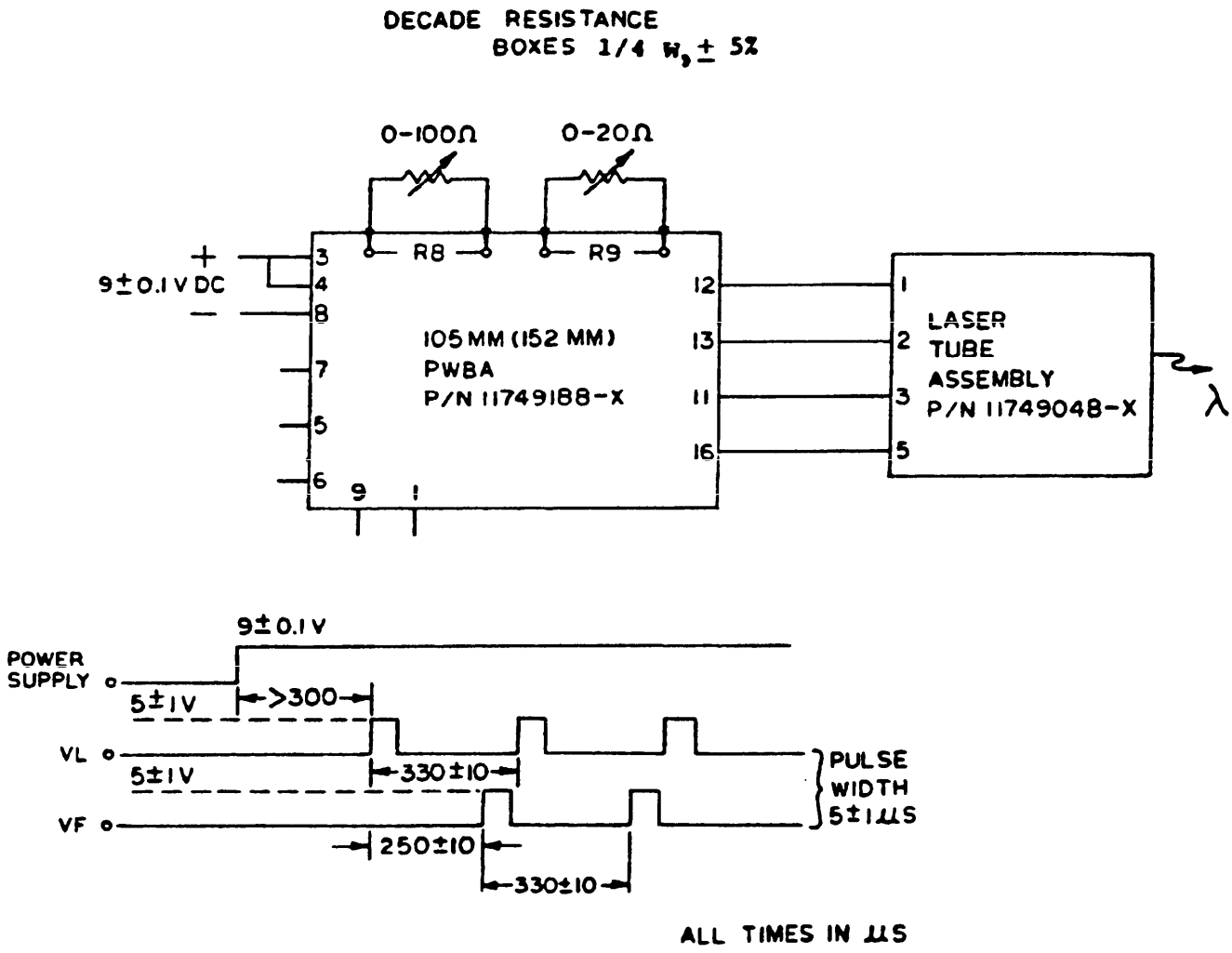


FIGURE 1. 105mm Laser Tube/Modulator Assy Power Source and Loads.

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3.4.2.1 Current drain. With power applied to terminals 3(+) and 8(-), the quiescent current drain of the laser tube/modulator assembly shall be less than or equal to 3mA.

3.4.2.2 Near miss energy (105mm and 152mm COAX). With power applied to terminals 3(+) and 8(-), terminal 6 connected to V_F, terminal 7 connected to V_L and terminals 5 and 9 connected, the near miss energy output shall be between 1.2 and 2.2 ergs.

3.4.2.3 Irradiance energy (105mm and 152mm COAX). With power applied to terminals 3(+) and 8(-), terminal 5 connected to V_L, terminal 6 connected to V_F and terminal 7 connected to terminal 9, the laser energy shall be within the 50 meter irradiance energy values given in Figure 2 at room temperature.

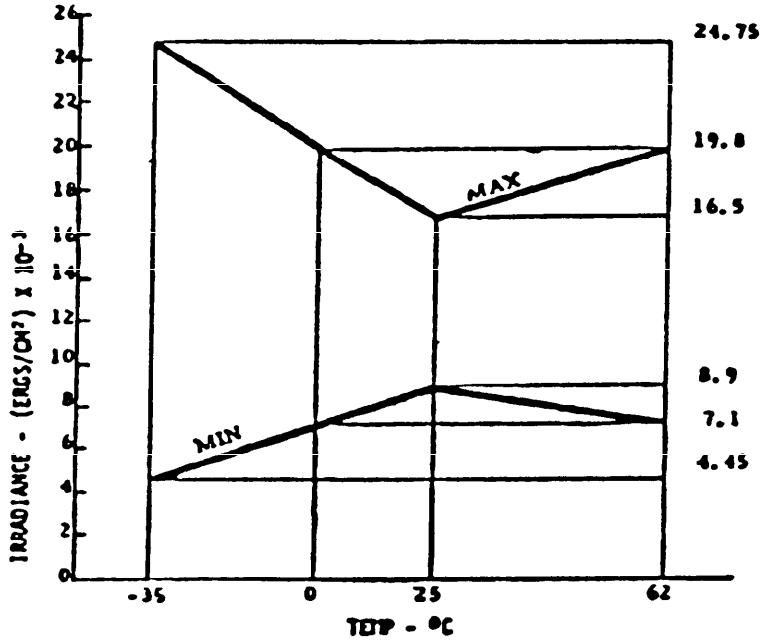


FIGURE 2. 50 Meter Beam Center Irradiance (COAX).

3.4.2.4 Near miss energy (105mm Main Gun). With power applied to terminals 3(+) and 8(-), terminal 6 connected to V_F, terminal 7 connected to V_L and terminal 5 connected to terminal

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9, the near miss energy output shall be greater than 1.2 times the total kill energy (per laser tube assembly data) but not more than 5.2 ergs.

3.4.2.5 Irradiance energy (105mm Main). With power applied to terminals 3(+) and 8(-), terminal 5 connected to V_L, terminal 6 connected to V_F and terminal 7 connected to terminal 9, the laser energy shall be within the 50 meter irradiance energy values given in Figure 3 at room temperature.

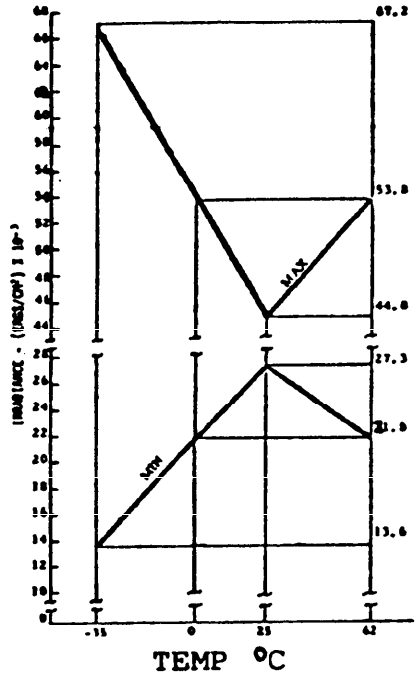


FIGURE 3. 50 Meter Beam Center Irradiance (Main Gun).

3.3.2.6 Laser pulses. With power applied to terminals 3(+) and 8(-), terminal 6 connected to V_F, terminal 5 connected to V_L and terminal 7 connected to terminal 9, the laser tube shall output pulses corresponding to the V_F pulses in frequency, width, and timing.

3.4.3 Environmental requirements. Unless otherwise specified the assembly shall be subject to the following environmental conditions and meet the stated requirements.

3.4.3.1 High temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2.

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when exposed to and after the following.

3.4.3.1.1 Operating and storage temperature. The assembly shall operate up to a temperature of +62.8°C (+145°F) after 4 hours of exposure at 62.8°C (145°F).

3.4.3.2 Low temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 when exposed to and after the following.

3.4.3.2.1 Operating temperature. The assembly shall operate down to a temperature of -31.7°C (-25°F) after exposure at -31.7°C for 4 hours.

3.4.3.2.2 Storage temperature. The assembly shall operate after exposure to a temperature of -34.4°C (-30°F) for a period of 24 hours.

3.4.3.3 Shock. The assembly shall meet the performance requirements of 3.4.2 after shocks with a nominal level of 40 g's terminal peak and nominal duration of 11 milliseconds as shown in Figure 4. Apply three shocks (18 total) in each direction of the three mutually perpendicular axis.

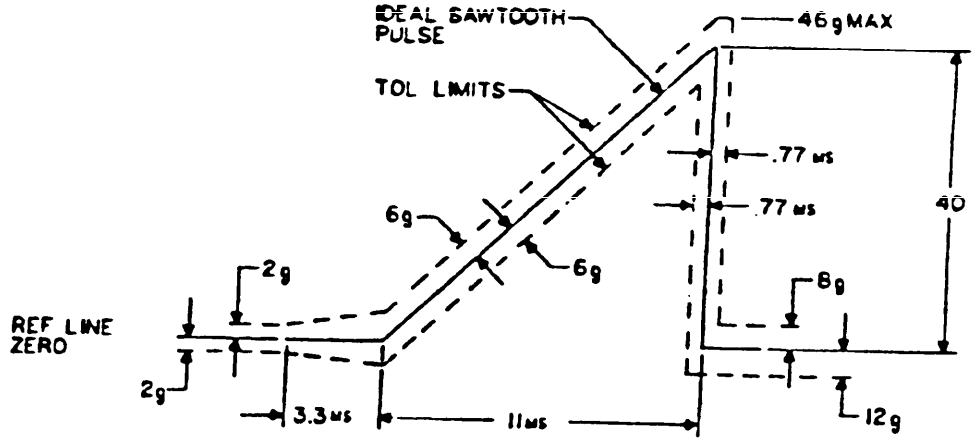


FIGURE 4. Shock Pulse

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3.4.3.4 Vibration. The assembly shall meet the performance requirements of 3.4.2 after sinusoidal vibrations of frequency and amplitude given below and in Figure 5 on three mutually perpendicular axes. Total duration shall be 60 minutes per axis.

Frequency	Amplitude	Sweep rate
7-30 Hz	±1.5 g	7-500-7 Hz
30-50 Hz	0.033" DA	in 15 minutes
50-500 Hz	±4.2 g	

3.5 Design and construction. The laser tube/modulator assembly shall be fabricated and assembled in accordance with drawings, parts lists, and other documents specified on drawing 11749188 and 11749048. The laser tube/modulator assembly shall be manufactured and tested in a working environment in accordance with Table I, work area 5 of MIL-STD-1695.

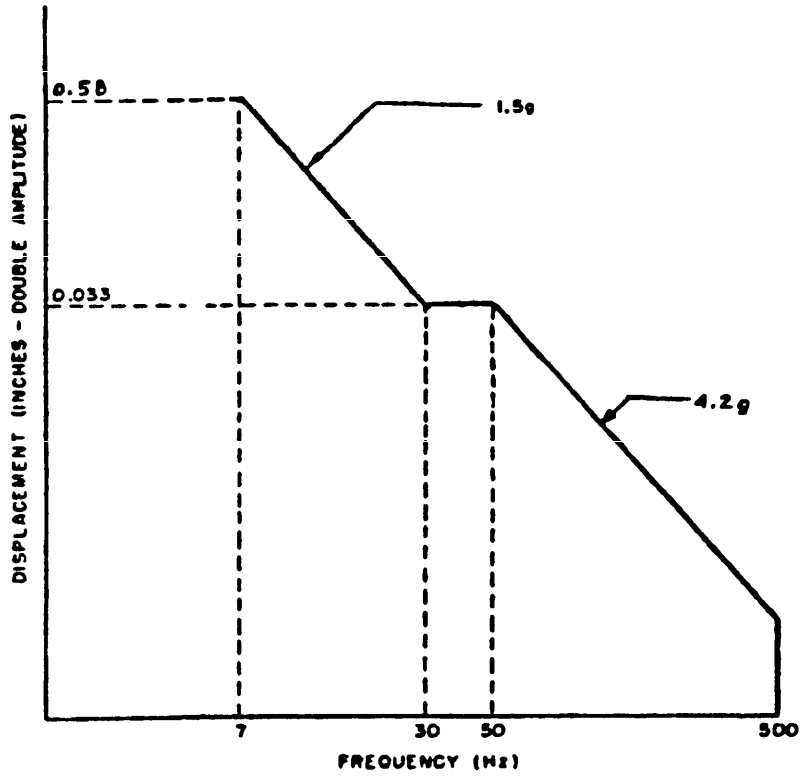


FIGURE 5. Vibration Test.

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3.5.1 Standards of manufacture. Handling of Electrostatic Sensitive Parts. This equipment contains semiconductor devices such as CMOS, mosfets, and microelectronic devices which may be degraded or damaged by an inadvertent discharge of static electricity. To prevent damage to these components, the supplier shall provide for adequate handling procedures and equipment in accordance with MIL-STD-1686 and DoD-HBK-263.

3.5.2 Soldering. Soldering shall be in accordance with the general requirements of section 4 of MIL-STD-2000.

3.5.3 Workmanship. General workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

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b. Environmental stress screen (see 4.5).

c. Quality conformance inspection (see 4.6).

d. Packaging inspection (see 4.6.4).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the working environment specified in 3.5 and at standard ambient conditions defined in 3.4.2 using the methods of 4.8 and 4.9.

4.4 First article inspection.

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of three printed wiring board assemblies.

4.4.2 Inspections to be performed. The first article inspections are listed in Tables IV and V. Each first article sample unit shall be inspected in accordance with the Table IV before being subjected to the environmental tests in Table V in the sequence listed. As determined by the Government, the first article assemblies may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

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TABLE IV. Examinations and Performance Tests. 1/ 2/

Class	Characteristics	Requirements Paragraph	Test/ inspection Method	First Article Sample Size	Quality 3/ Conformance Inspection Sample Size/ Accept Criteria
101	Current Drain	3.4.2.1	4.8.2.1	3	32/Acc-0,Rej-1
102	Near Miss Energy (COAX)	3.4.2.2	4.8.2.2	3	32/Acc-0,Rej-1
103	Irradiance Energy (COAX)	3.4.2.3	4.8.2.3	3	32/Acc-0,Rej-1
104	Near Miss energy (Main Gun)	3.4.2.4	4.8.2.4	3	32/Acc-0,Rej-1
105	Irradiance Energy (Main Gun)	3.4.2.5	4.8.2.5	3	32/Acc-0,Rej-1
106	Laser Pulses	3.4.2.6	4.8.2.6	3	32/Acc-0,Rej-1
201	Workmanship	3.5.3	Visual	3	125/Acc-0,Rej-1
202	Soldering 4/	3.5.2	Visual	3	125/Acc-0,Rej-1
203	Conformal Coating	3.5	Visual	3	125/Acc-0,Rej-1
204	Dim Characteristics	3.5	SME 5/	3	125/Acc-0,Rej-1
205	Comp Orientation	3.5	Visual	3	125/Acc-0,Rej-1
206	Comp and Board Identification	3.5	Visual	3	125/Acc-0,Rej-1

NOTES:
1/ ESS testing per 4.5 is required for all deliverable units.
2/ 200 class (minor characteristics) examinations required prior to 100 class (major characteristics) performance tests.
3/ For lot sizes less than 32 or 125 as listed, 100% inspection is required (see 4.6.5).
4/ Check before conformal coating is applied.
5/ Standard Measuring Equipment.

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TABLE V. Environmental Tests.

Class	Characteristic	Requirements Paragraph	Test Paragraph	First Article Sample Size	Quality Conformance Inspection Sample Size, Accept Criteria
107	High Temp	3.4.3.1	4.9.1	3	1/Acc-0, Rej-1
108	Low Temp	3.4.3.2	4.9.2	3	1/Acc-0, Rej-1
109	Shock	3.4.3.3	4.9.3	3	1/
110	Vibration	3.4.3.4	4.9.4	3	1/

NOTES:
1/ Not required for quality conformance inspection.

4.4.3 Rejection. If any assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly to comply with any of the requirements.

4.5 Environmental stress screen. An environmental stress screening program shall be established. All units shall be subjected to a thermal stress screen in accordance with Table VI and the procedure below.

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TABLE VI. ESS Temperature Cycling Requirements.

Low Temp	High Temp	Rate Change	Power	Monitor	Cycles/ Group A Test
-34.4°C	+62.8°C	15°C/minute	Off	No	20/Failure Free or Additional Cycle

4.5.1 Conformal coating. Thermal stress screen be shall be performed prior to conformal coating unless another method is approved by the Government.

4.5.2 Chamber temperature. Chamber temperature cycling shall be from -34.4°C to +62.8°C (-30°F to +145°F) and the duration of one cycle shall be 8 minutes up and 8 minutes down plus 4 minutes dwell at high temperature and 4 minutes at low temperature for a total thermal cycle duration of 24 minutes. The units shall begin and end the cycles at ambient temperature (see Figure 6).

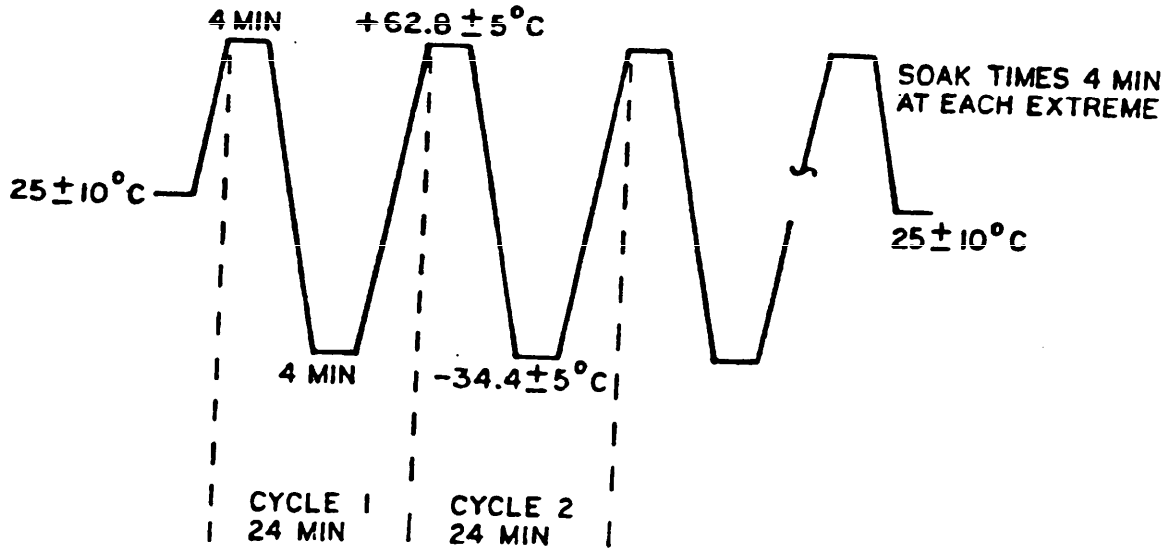


FIGURE 6. Environmental Stress Screen Temperature Profile.

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4.5.3 Cycle frequency. A total of twenty temperature cycles are to be imposed on each unit.

4.5.4 Temperature cycles. Temperature cycling shall be conducted with units nonoperating and without power (static).

4.5.5 Test requirements. After the thermal stress screen, the units shall meet the Group A tests (4.6.2). If a unit is rejected, the unit shall be repaired for the failure (if possible) and be put through one additional cycle of ESS. The unit shall then undergo and pass the Group A tests.

4.6 Quality conformance inspection. Quality conformance inspection requirements are defined in 4.6.2 and 4.6.3.

4.6.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn of which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.6.2 Group A tests. Group A tests consist of examinations and performance tests listed in Table IV in the quantities specified.

4.6.3 Group B tests. Group B tests shall consist of the tests indicated in Table V and shall be conducted on one sample unit per inspection lot which has been subjected to, and passed, Group A tests (see 4.6.2).

4.6.4 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall verify that the units are packaged in accordance with the requirements of 5.1. During the period of first article inspection, one

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additional unit, packaged in accordance with 5.1, shall be examined to the requirements for quality conformance inspection listed below. Failure of packaging shall be cause for rejection of packaging. Two units per lot shall be examined for the following:

a. Verify that the outside bag is marked in accordance with MIL-STD-129 and is properly closed.

b. Verify that sleeve is around box and that foam properly surrounds inner bag.

c. Verify that the inner bag is properly sealed.

d. Verify that the unit is not damaged after the removal of all packaging materials.

4.6.5 Failure of sample. Should any one sample fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract (see 6.7).

4.7 Inspection equipment.

4.7.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607 and MIL-W-63150.

4.7.2 Contractor-furnished inspection equipment.

4.7.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAP/QARs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs (see 6.3).

4.7.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the Methods of Inspection specified in 4.8 and 4.9 of this specification and with the component inspection procedures specified in Examination and Test Facilities requirements of MIL-F-13926, whenever Government designs are not specified (see 4.7.2.1). Since tolerance of test equipment is normally considered to be within

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ten percent of product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. When specified in contract or purchase order, the contractor shall submit inspection equipment designs for government review and evaluation (see 6.3).

4.7.2.3 Test equipment. In conjunction with 4.7.2.1 and 4.7.2.2, the test equipment listed in Table VII (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.8.

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in accordance with 5.1, shall be additional unit packaged

TABLE VII. Test Equipment List. 1/

Power Supplies (DC)	Standard - capable of providing levels and limits stated in 4.8.1 and 4.8.2.
Ammeter/Multimeter	Standard - capable of measuring currents stated in 4.8.2.1
Resistance Decade Boxes	Standard - capable of providing ohmage stated in 4.8.1.1 to 4.8.1.2.
Dual Output Pulse Generator	Standard - capable of providing pulses stated in 4.8.1 and 4.8.2.
Laser Energy Meter and Pulse Integrator	Equivalent to EG&G Model 460-1A and 460-3 - capable of measuring laser energy specified in 4.8.1 and 4.8.2.
Laser Detector Head	Equivalent to EG&G Model 460-2 - capable of detecting laser energy specified in 4.8.1 and 4.8.2.
Radiometer/Photometer and Pulse Integrator	Equivalent to EG&G 550-1 and 550-3 - capable of measuring irradiance laser energy specified in 4.8.1 and 4.8.2.
Silicon Multiprobe	Equivalent to EG&G 550-2 - capable of detecting laser energy specified in 4.8.1 and 4.8.2.
Si-PN Detector Module	Part Number 11748854 - used in 4.8.2.6.
Helmet (MWLD) PWBA	Part Number 11749020 - used in 4.8.2.6.
Dual Channel Oscilloscope	Standard - capable of measuring pulses in 4.8.2.6.

NOTE: 1/ Must be calibrated to the requirements of MIL-STD-45662 prior to use.

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4.8 Methods of inspection. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the Government procuring agency for approval prior to use (see 4.7.2.2).

4.8.1 Matched assemblies. To obtain the matched assemblies, connect each laser tube/modulator assembly into the circuit of Figure 7 for the purpose of identifying resistor values R8 and R9 of the assemblies. In order to meet the requirements of 3.1.1, apply power by turning ON S1 and leaving all other switches as shown unless otherwise specified. Set the pulse generator to provide the V_L and V_r pulses shown in Figure 1. Turn on the laser power meter, set the power meter and detector head to 904nm, and set the decade range on the integrator module to 10^{-6}. Turn on the radiometer/photometer, turn on the autorange, set the unit selector thumbwheel and the multiprobe switch to microJoules/cm^2 (note that 1 erg = 10^{-7} Joule) and set the decade range on the integrator module to 10^{-3}.

4.8.1.1 Setting R8. Set S2 to position M, turn ON S3, set R9 to a nominal value from the range given in 3.1.1, align the power meter and energy detector with the laser and adjust R8 so that the appropriate near miss energy of 3.1.1 is indicated on the power meter. Replace R8 with a fixed resistor value from PL 11749188 nearest in value.

4.8.1.2 Setting R9. Set S2 to position K, turn ON S3, align the radiometer and multiprobe with the laser and adjust R9 so that the appropriate irradiance energy of 3.1.1 is indicated on the radiometer. Replace R9 with a fixed resistor given in PL11749188 nearest in value.

4.8.2 Performance tests. After R8 and R9 have been fixed the following tests and examinations shall be performed to demonstrate compliance with 3.4.2. The laser tube/modulator assembly shall be connected to the circuit specified in Figure 7. Apply power by turning ON S1 and leave on until all tests have been completed. All other switches shall remain as shown unless otherwise specified in 4.8.2.1 through 4.8.2.6. Set the pulse generator to provide the pulses shown in Figure 1. Turn on the laser power meter, set the detector head and the power meter to 904 nm, and set the decade range on the integrator module to 10^{-6}. Turn on the radiometer/photometer, turn on the autorange, set the unit selector thumbwheel and the multiprobe switch to microJoules/cm^2 (note that 1 erg = 10^{-7} Joule) and set the decade

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range on the integrator module to 10^{-3}.

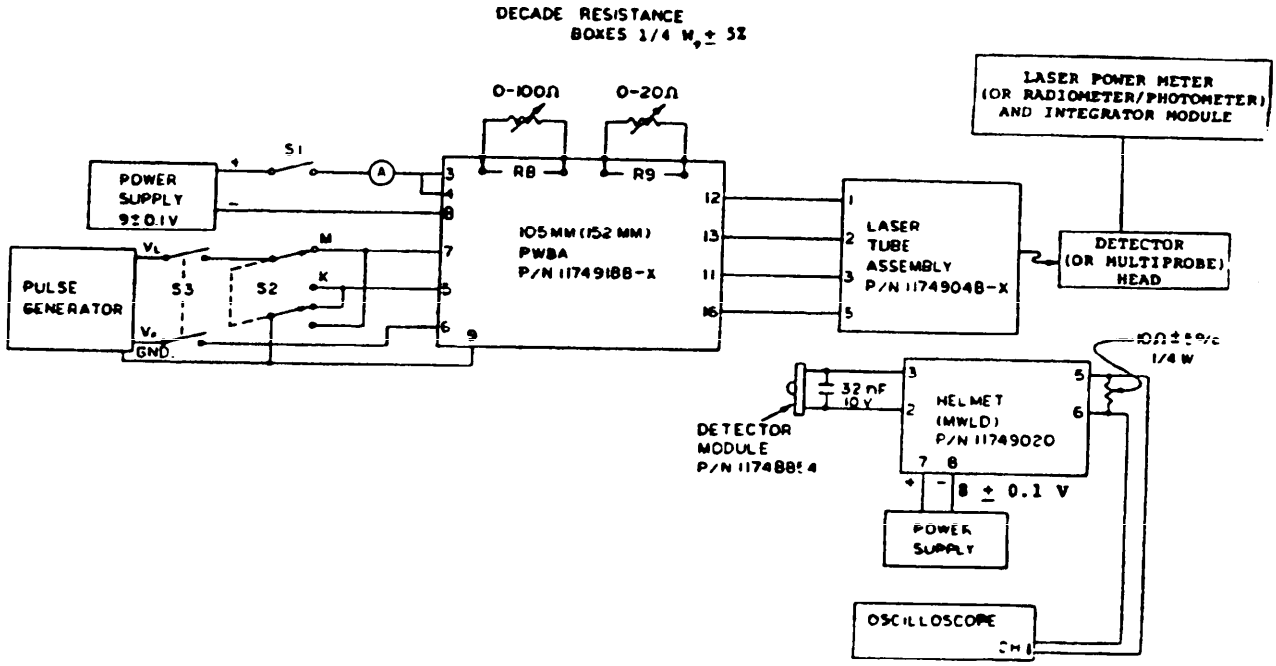


FIGURE 7. Test Circuit for 105mm, 152mm.

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4.8.2.1 Current drain. Turn ON S1 and verify that the quiescent current drain meets the requirements of 3.4.2.1.

4.8.2.2 Near miss energy (105mm and 152m COAX). Test the COAX near miss energy by setting S2 to position M, turning ON S3, aligning the laser detector with the laser and verifying that the energy requirements of 3.4.2.2 are met on the power meter.

4.8.2.3 Irradiance energy (105mm and 152mm COAX). Test the COAX hit energy by setting S2 to position M, turning ON S3, aligning the silicon multiprobe with the laser and verifying that the irradiance requirements of 3.4.2.3 are met on the radiometer.

4.8.2.4 Near miss energy (105mm Main Gun). Test the main gun near miss energy by setting S2 to position M, turning ON S3, aligning the laser detector with the laser and verifying that the energy requirements of 3.4.2.4 are met on the power meter.

4.8.2.5 Irradiance energy (105mm Main Gun). Test the main gun hit energy by setting S2 to position K, turning ON S3, aligning the silicon multiprobe with the laser and verifying the irradiance requirements of 3.4.2.5 are met on the radiometer.

4.8.2.6 Laser pulses. Test the laser pulses by replacing the radiometer and power meter with the detector module and helmet (MWLD) combination. Power the helmet (MWLD) and the oscilloscope, set S2 to position K, turn ON S3 and verify the requirements of 3.4.2.6 are met by making measurements with the oscilloscope.

4.9 Environmental tests.

a. Unless otherwise specified, environmental tests shall be conducted in accordance with the test conditions and general test performance guidance of MIL-STD-810. Test facilities, chambers and apparatus used in conducting the test required by this specification shall be capable of meeting the conditions specified.

b. Where necessary in the following tests, it shall be permissible to temporarily open chamber doors to permit access to and manipulation of test equipment and sample.

4.9.1 High temperature. This satisfies the storage and operating conditions of high temperature requirements as specified in 3.4.3.1.

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4.9.1.1 Storage and Operating. The assembly shall be tested in accordance with MIL-STD-810, Method 501 using the following procedure:

a. Step 1 - Place the test item, configured for high temperature test, into the temperature chamber.

b. Step 2 - Raise the internal temperature to 62.8°C (145°F).

c. Step 3 - Maintain the internal chamber temperature for a period of 4 hours while insuring the relative humidity is not in excess of 15 percent.

d. Step 4 - Operate the test item until the item is stabilized and perform the tests required in 4.8.2.

e. Step 5 - Return the test item, nonoperating, to standard ambient condition and stabilize.

f. Step 6 - Visually inspect for physical damage, and in the absence of physical damage, perform the tests required in 4.8.2.

4.9.2 Low temperature. This satisfies the storage and operating conditions of low temperature requirements as specified in 3.4.3.2.

4.9.2.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 502 using the following procedure:

a. Step 1 - Place the test item, configured for low temperature test, into the temperature chamber.

b. Step 2 - Lower the internal chamber temperature to -34.4°C (-30°F) and maintain for a period of 24 hours after stabilization.

c. Step 3 - Adjust the internal chamber temperature to -31.7°C (-25°F) and maintain until temperature stabilization of the test item is reached.

d. Step 4 - Operate the test item until the item is stabilized and perform the tests required in 4.8.2.

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e. Step 5 - Return the test item, nonoperating, to standard ambient conditions and stabilize.

f. Step 6 - Visually inspect for physical damage, and in the absence of physical damage, perform the tests required in 4.8.2.

4.9.3 Shock. The assembly, when mounted on a fixture that simulates its next higher assembly (see Figure 8), shall be tested in accordance with MIL-STD-810, Method 516.3, Section II, Procedure I, with the following modifications:

a. Step 3 - the assembly shall be in a non-operational mode.

b. The step 3 test transient shall be in accordance with 3.4.3.3.

c. Step 5 - the assembly inspection and functional verification shall not be conducted until shock has been completed in all three axes.

d. After shock testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.2.

4.9.4 Vibration. The assembly, when mounted on a fixture that simulates its next higher assembly (see Figure 8), shall be tested in accordance with MIL-STD-810, Method 514.3, Section II, Procedure I, for a duration of 60 minutes with the following modifications:

a. Step 4 - mount the test item on vibration equipment.

b. Step 5 - expose the assembly to test level and durations as specified in 3.4.3.4.

c. Step 6 - does not apply.

d. Step 7 - the assembly inspection and functional verification shall not be conducted until vibration has been completed in all three axes.

e. After vibration testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.2.

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4.10 Certifications. Certification shall be provided that all units have been produced in accordance with the requirements of 3.5 and packaged in accordance with the requirements of 5.1.

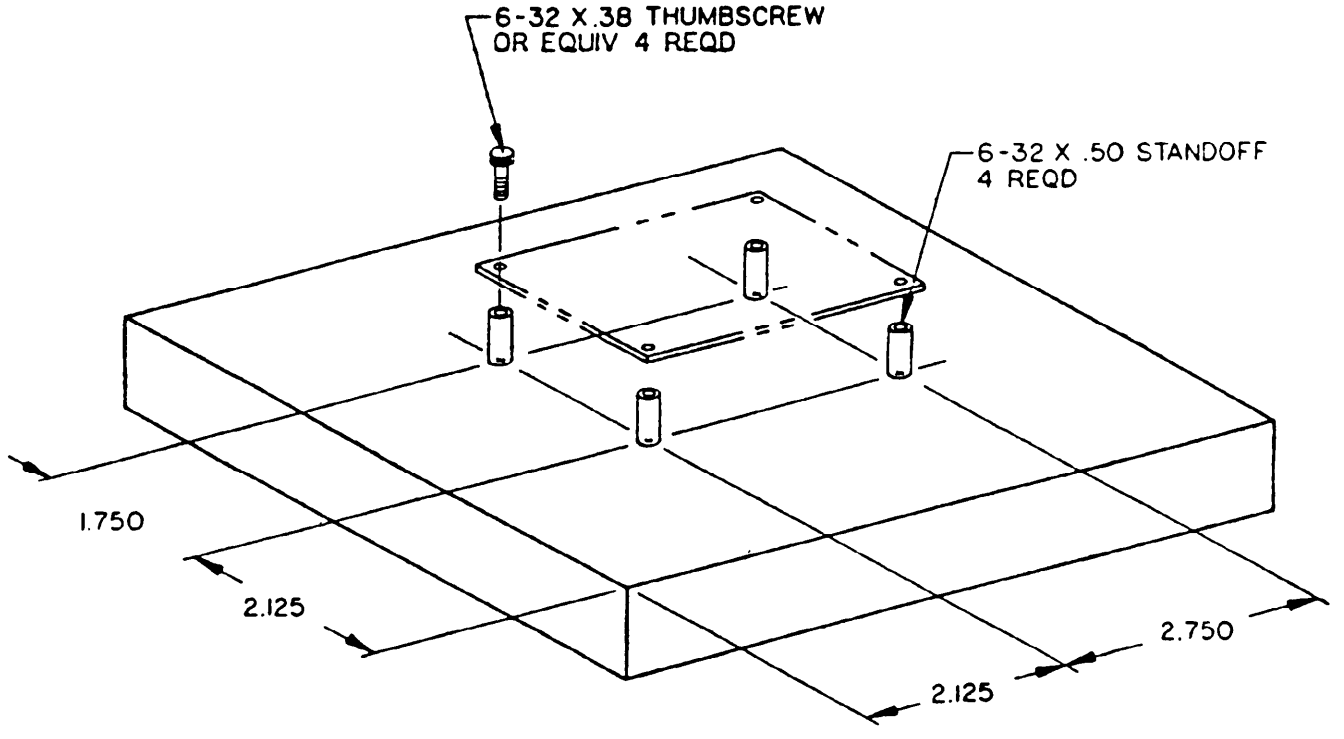


FIGURE 8. Mounting Configuration.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI-P11835692, SPI-P11749644 and SPI-P11835692. The level of protection shall be A, B, or C, as defined by the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The following laser/tube modulator assemblies are part of the Multiple Integrated Laser Engagement System: 105mm COAX, P/N 11835692; 152mm COAX, P/N 11835693; and 105mm Main Gun, P/N 11749644.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of this specification (see page 1).

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Special tests and examinations required (see 4.2); the type of inspection required; and the activity responsible for testing, and its facility (see 4.1).

d. Whether a first article sample is required (see 3.2), and if so, pertinent details (see 4.4) including disposition of the first article sample.

e. Certification of conformance for each lot or shipment of product.

f. Whether Government-Loaned Property (Government Furnished inspection/test equipment) of 6.8 is available or will the contractor be required to furnish inspection equipment (see 4.7).

g. Applicable level(s) of preservation, packaging, packing and marking instructions (see 5.1).

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6.3 Submission of contractor inspection equipment designs for approval. Unless otherwise specified, submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. The letter of submittal shall specify, the contractor; contract number; title, number and revision of this specification; item nomenclature; classification of defect and test paragraph.

6.4 Definitions.

6.4.1 Hit. Successful transmission of the weapon simulator kill code that results in a "hit" on the target, but that does not "kill" the target. An audio alarm at the target signifies the hit.

6.4.2 MILES. Multiple Integrated Laser Engagement System.

6.4.3 Near Miss. Illumination of the target with the near miss laser message resulting in an audio indication to the target that it is under fire.

6.4.4 SME. Standard Measuring Equipment.

6.5 Drawings. Drawings listed in Section 2 of this specification under heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group

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A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

6.7 Government-loaned property. Unless otherwise specified (see 6.2), the contracting officer should arrange to loan the following Government furnished Inspection/Test equipment specified below:

P/N	Title	Ref. Paragraph
11748854	Detector Module, Si-PN	4.8.2.6
11749020	PWBA, Helmet (MWLD)	4.8.2.6

6.8 Subject term (key word) listing.

Detector Module
Fire Control
Training Devices
Transmitter

Custodians:
Army - AR

Preparing activity:
Army - AR

(Project 1265-A011)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisio letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization t waive any portion of the referenced document(s) or to amend contractual requirements.

1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
MIL-P-70847 (AR)	23 January 1992

3. DOCUMENT TITLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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